TTT4295 - Acoustic Signal Processing
Exercise 2 - Microphone arrays.

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All tasks, except those marked *Optional* are to be handed in for approval (online, in blackboard). The deadline is specified in blackboard.

Tasks

1. **Minimalistic array: Two omni microphones**
   Two omnidirectional microphones are positioned as illustrated in figure 1, where a plane wave is coming from the angle θ.

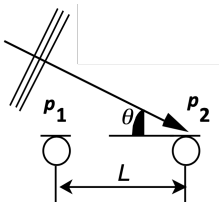


Figure 1: Two omnidirectional microphones, and an incident plane wave.

a) What is the relationship between the sound pressures p_1 and p_2, at the two microphones, for the incident plane wave? That is, what is the ratio p_2/p_1 for the incident plane wave, where p_1 and p_2 are the complex amplitudes for a harmonic plane wave of (angular) frequency ω?

b) Assuming that the distance is much smaller than the wavelength of the incident plane wave, what is the simplified relationship between the two sound pressure amplitudes?

c) In figure 2, the two blocks A and B indicate some kind of filters. Show that a cardioid microphone can be implemented by reversing the polarity in one of the two blocks (A,B), and implementing a delay Δt in the other. A cardioid microphone should have an output signal e_{summed}, after summation, that shows a cardioid-shaped directivity, $e_{summed} \propto p_{\text{in}}(1+\cos\theta)$,

where p_{in} is the sound pressure amplitude for the incoming wave. Assume that:
• the output signal of each microphone is proportional to the sound pressure at that microphone,
• the distance L is small compared to the wavelength λ,
• the output signal doesn't need to have flat frequency response.

What value should Δt have?

d) In order to get a flat frequency response for the output signal e_{summed}, what extra filter needs to be inserted?

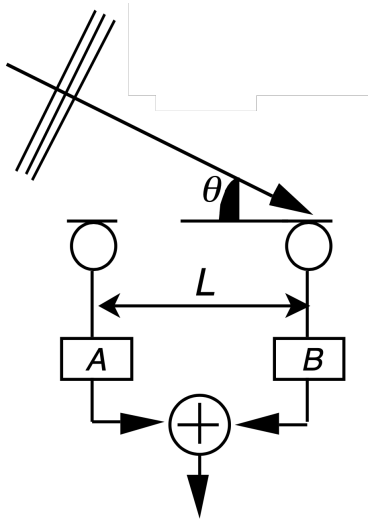


Figure 2: Two omnidirectional microphones, the output signals of which are filtered and summed, to syntheisze a single diretional microphone.

2. **The directivity index of a simple second-order microphone**

In this task, you should derive the Directivity factor, DF, and Directivity index, DI, for one particular type of *second-order microphone*. The microphone has this directivity function:

$$D(\theta) = \cos(2\theta)$$

Remember the definitions of DF and DI, for a rotationally symmetrical directivity function, $D(\theta)$ which is normalized such that $D(0) = 1$::

$$DF = \frac{1}{0.5 \int_0^{\pi} |D(\theta)|^2 \sin\theta d\theta}$$

and

$$DI = 10 \lg DF$$

3. (*Optional*) **More general second-order microphone**

   Now, for a more general second-order microphone, with

   $$D(\theta) = a + b\cos(2\theta)$$

   To normalize this microphone, the parameter a must equal $1 - b$.

   Task 1: For this microphone, show that the expression for the DF equals

   $$DF = \frac{1}{1 - \frac{8}{3}b + \frac{32}{15}b^2} \tag{1}$$

   Task 2: For the expression in Eq. (1), derive which value of b gives the highest DF value.

   Task 3: What is the highest possible DI, in dB, for this microphone?

4. **Combining three microphones**

   We have a special type of first-order microphone, with the directivity function

   $$D(\theta) = 0.6 + 0.4\cos\theta$$

   We use three of these microphones and let them point in the directions $0°, 120°$, and $240°$. We want to mix/add these three microphone signals together to get three virtual microphones with these three wanted directivity functions:

   $$D_W(\theta) = 1, \quad D_X(\theta) = \cos\theta, \quad D_Y(\theta) = \sin\theta$$

   How should the three microphone signals be combined to give these three virtual directivity functions?

   Hint: Introduce the angle $\theta' = \theta - 120°$, and express the directivity function for the microphone pointing in the direction of $\theta = 120°$ as $D_2(\theta') = 0.6 + 0.4\cos\theta'$ so that $D_2(\theta) = 0.6 + 0.4\cos(\theta - 120°)$. Do the same for the third microphone, using $D_3(\theta'')$ with $\theta'' = \theta - 240°$. Once all three microphones have their directivity functions expressed as function of θ, add them together as

   $$D_{\text{virtual}}(\theta) = w_1 D_1(\theta) + w_2 D_2(\theta) + w_3 D_3(\theta)$$

   and choose w_1, w_2, and w_3 such that the virtual microphone gets the desired directivity function.